



Computer Applications Notes-1

information communication technology (Kenya Medical Training College)



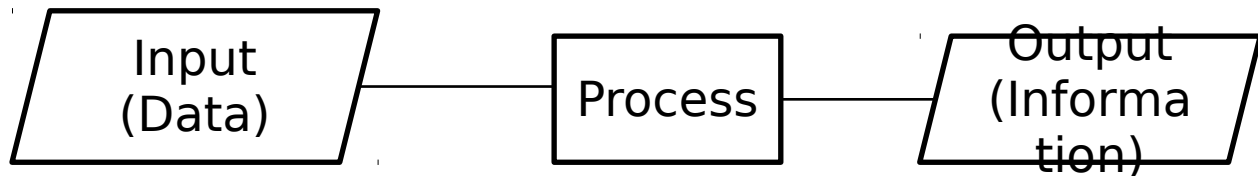
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INTRODUCTION TO COMPUTERS

- Information and communication plays an important role in our day to day activities.
- It has become a common practice to find information based systems in banks, retail stores, government agencies, entertainment industry and media houses.
- To adapt to the ever changing technological world, one needs to be a knowledgeable and competent user of computer based technologies.

DEFINATION OF TERMS

- **Computer**
 - A computer can be defined as an electronic device that processes a users input usually referred to as data to a desired output ,also known as information.
 - This is made possible by a set of instructions called **computer programs** stored in a computer.



Input, Process and Output

- **Data** can be defined as raw facts that do not have much meaning to the user and may include; numbers, letters and symbols.
- **Information** is the processed data that is meaningful to the user.

A computer is said to be **electronic** because it utilizes electrical signals to process information.

- **Information Communication and Technology (ICT)**
 - Is the integration of computers and telecommunication facilities for the purpose of communication.
- Computers come in different sizes and designs but the most common is referred to as a Personal Computer (PC).
- Personal computers are mostly used in offices, schools, business premises and at home.

Physical Parts of a Computer

- A desktop computer is made up of a collection of different components that are interconnected in order to function as a single entity.
- A typical desktop computer is basically made up of a system unit and other devices connected to the system unit called **peripheral devices**.
- Examples of peripheral devices include:
 - Monitor/Screen
 - Keyboard
 - Mouse

System Unit

- This is the part that houses the brain of the computer called the Central Processing Unit (CPU).
- The system unit also houses other devices called **drives**.
- Drives are used to store, record and read data.
- The two common types of system units(form factors) are:
 1. Tower Type
 2. -Desktop Type

Peripheral Devices

- Peripheral devices are connected to the system unit using special cables called **interface cables** that transmit data and information to and from the devices.
- The cables are attached to the system unit through connectors called **ports**.
- Peripheral devices may be arranged with the monitor resting on top of the system unit or a stand alone if connected to a tower type system unit.

Keyboard

- A keyboard is the most common peripheral device that enables a user to enter data and instructions in a computer.

Mouse

- A mouse is a pointing device, used to execute commands.
- It controls an arrow displayed on the screen.

Monitor

- A **monitor** or simply the **screen** is a television like device used for displaying information.
- It is called a monitor because it enables the user to monitor what is going on in the computer.

CLASSIFICATION OF COMPUTERS

- Computers can be classified according to:
 - Physical Size.....i.e. supercomp,mainframe etc
 - Purpose.....i.e. general purpose and special purpose comps.
 - Functionalityi.e. analogue, digital and hybrid comps
 - Generationi.e. 1st generation,etc

Classification According to Physical Size

- Based on physical size, computers can be classified into four main groups namely:
 - Supercomputers
 - Mainframe Computers
 - Mini Computers
 - Micro Computers

Supercomputers

- Supercomputers are the fastest, largest, most expensive and powerful computers available.
- They are able to perform many complex operations in a fraction of a second.
- Because of its weight, a supercomputer is kept in a special room.

- Due to their huge processing power, supercomputers generate a lot of heat.
- Special cooling systems are therefore required.
- Sometimes the whole **CPU** is immersed in an **aquarium like tank** containing liquid **fluorocarbon** to provide cooling.
- Supercomputers are mainly used for scientific research, which requires enormous calculations.
- Applications that justify the use of supercomputers include: **Aerodynamic design and simulation, petroleum research, defense and weapon analysis** among others.
- Supercomputers are mainly found in developed countries.

Mainframe Computers

- Mainframes are less powerful and less expensive than supercomputers.
- They are used for processing data and performing complex mathematical calculations.
- They have a large storage capacity and can support a variety of peripherals.
- They are mostly found in government agencies, big organizations and companies such as banks, hospitals, airports etc. which have large information processing needs.

Minicomputers

- Resembles the mainframe but is slightly smaller.
- It is not as powerful and fast as the mainframe computer.
- It was developed as a cheaper alternative to the mainframe computer.
- They are used mainly in scientific laboratories, research institutions, engineering plants and places where processing automation is required.

They are well adapted for functions such as accounting, word processing, database management and specific industry applications

Micro Computers

- A microcomputer is the smallest, cheapest and relatively least powerful type of computer.
- It is called a micro computer because its CPU is called a *microprocessor*.
- Microcomputers are commonly used in training and learning institutions, small business enterprises, and communication centers among others.

- Technological advancement has seen the development of smaller microcomputers. The various types include:
 - Desktop Computer
 - Notebook or Laptop Computer
 - Personal Digital Assistant (PDA)

Classification According to Purpose

- Computers can be classified according to the tasks they perform either as **general** or **special purpose** computers.

General Purpose Computers

- They have been designed to be able to perform a variety of tasks when loaded with appropriate programs.
- They are the most common types of computers in use today.
- Their flexibility enables them to be applied in a wide range of applications like document processing, performing calculations, accounting etc.

Special Purpose Computers

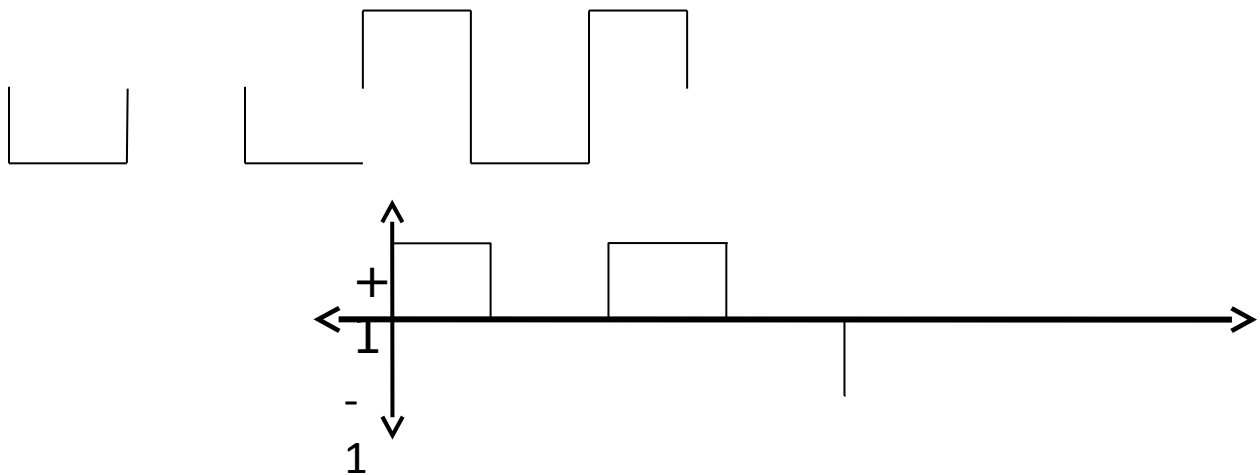
- Are designed to serve a specific purpose or to accomplish one particular task.
- Such computers can perform no other task except the one they were meant to do.
- Examples of such computers include: Robots used in manufacturing industries, Mobile phones for communication only and electronic calculators that carry out calculations only.
- Since special purpose computers are dedicated to a single task, they can perform the task quickly and very efficiently.

Classification According to Functionality

- Computers can be classified into three types according to the type of data they process.
- Data can either be in **discrete** or in **continuous** form.
- They are classified as:
 - Digital Computers
 - Analog Computers
 - Hybrid Computers

Digital Computers

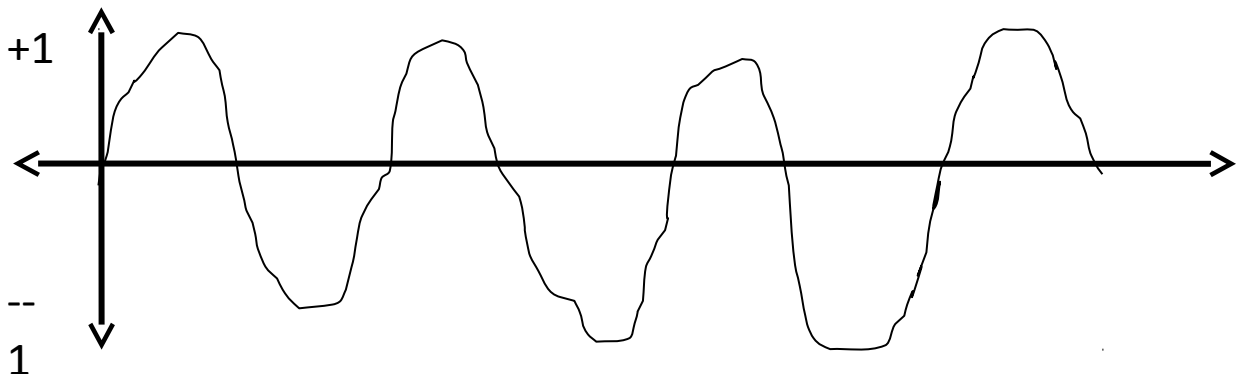
- They process data that is discrete in nature.
- Discrete data also known as digital data is usually represented using a two state square waveform.



Analog Computers

- They process data that is continuous in nature.

Continuous data also known as analog data is usually represented using a continuous waveform



Hybrid Computers

- Hybrid Computers are designed to process both analog and digital data.

Classification of Computers according to Generations

- The view of computers into generations is based on the fundamental technology employed.
- Each new generation is characterized by greater speed, larger memory capacity and smaller overall size than the previous one.

1ST Generation Computers (1946 – 1957)

- Used Vacuum Tubes or Thermionic valves to store and process information.
- Used Magnetic Drum Memories
- Used punch Cards to enter data.
- Maximum memory was 2KB.
- They were very slow - their speed was measured in **Milliseconds**.
- These computers were large in size and writing programs on them was difficult.
- The following are major drawbacks of First generation computers.
 - The operating speed was quite slow.
 - Power consumption was very high.
 - It required large space for installation.
 - The programming capability was quite low.
 - Cumbersome to operate – switching between programs, input and output

2ND Generation Computers (1958 - 1964)

- Replaced vacuum tubes with Transistors.

- The transistor is smaller, cheaper and dissipates less heat than a vacuum tube.
- Consumed less power
- Faster than 1st generation.
- Used **Magnetic Core Memories**
- Their speeds were measured in **Microseconds**.
- **Ram MEMORY** size expanded to 32KB
- The second generation also saw the introduction of more complex arithmetic and logic units, the use of **High – Level Programming Languages** and the provision of system software with the computer.
- Transistors are smaller than electric tubes and have higher operating speed. They have no filament and require no heating. Manufacturing cost was also lower. Thus the size of the computer got reduced considerably.
- It is in the second generation that the concept of Central Processing Unit (CPU), memory, programming language and input and output units were developed.

3RD Generation Computers (1965 - 1971)

- Had an Integrated Circuit called Small Scale Integration (SSI)
- were smaller in size
- lower cost
- larger memory and processing speed was much higher
- The transistors were individually mounted in separate packages and interconnected on printed circuit boards by separate wires. This was a complex, time consuming and error-prone process.
- Although the transistor technology was a major improvement over vacuum tubes, problems remained.

The early integrated circuits are referred to as small-scale integration (SSI). Computers of this generation were smaller in size, lower cost, larger memory and processing speed was much higher

4TH Generation Computers (1972 - Present)

- Employ Large Scale Integrated (LSI) and Very Large Scale Integrated (VLSI) circuit technology to construct computers.
- Over 1,000 components can be placed on a single integrated-circuit chip.

5TH Generation Computers

- These are computers of 1990s
- Use Very Large Scale Integrated (VLSI) circuit technology to build computers.
- Over 10,000 components can be incorporated on a single integrated chip.
- The speed is extremely high in fifth generation computer.
- Apart from this it can perform *parallel processing*. The concept of *Artificial intelligence* has been introduced to allow the computer to take its own decision.

COMPUTER SYSTEM COMPONENTS

- A computer system is a collection of three entities namely;
 - Hardware
 - Software
 - Live ware that work together to receive, process, manage and present information in a meaningful format.
- **Hardware:** Physical or tangible components that make up a computer system.
- They are classified into four categories namely:
 - input devices
 - central processing unit
 - output devices
 - Storage devices.
- **Software:** Refers to a set of instructions that direct a computer on what to do.

- They are mainly classified into two categories:
 - System software
 - Application Software.

liveware

-Is an informal term that refers to the human beings attachment to computers.

- Other terms used are:

Wetware

-is a slang that describes the human element of the Information Technology (IT).

Gray ware

-Is a classification of applications that behaves in a manner that is annoying or undesirable and troublesome than malware.

These may include users, people who make computer programs (programmers) and computer engineers.

Input Devices

- Input devices convert user input which is in human readable form to machine language that a computer can process.
- These devices can be classified according to the methods that they use to enter data, namely:
 - Keying devices
 - Pointing devices
 - Scanning Devices
 - Other technologies used to capture data.

Keying Devices

- Typing is the most common way to input data.
- A keying device converts typed numbers, letters and special characters into machine readable form before processing takes place.
- Types of keying devices include;

- Traditional Keyboards
- Flexible keyboards
- Ergonomic keyboard
- Keypad
- Braille keyboard

Pointing Devices

- Pointing devices are used for controlling a pointer cursor on the screen.
- Apart from the mouse other pointing devices include:
 - » Trackball
 - » Joystick
 - » Light pen and stylus

The different types of mouse include:

- » Standard mouse
- » Optical mouse
- Cordless Mouse

Scanning Devices

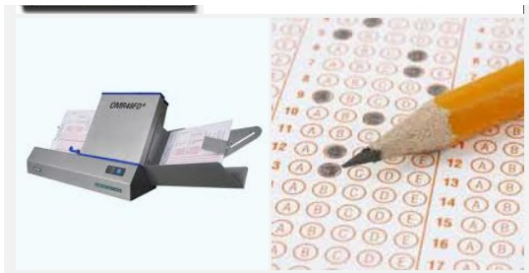
- Scanning means capturing data from an object and converting it into digital format.
- Scanning devices can be classified according to technology used to capture data.
- These are:
 - » Optical Scanners
 - Magnetic Scanner

1. OPTICAL SCANNERS

- Optical scanners capture data using optical or light technology.
- A light beam passes over an object and the image is analyzed by specialized software.
- **Some examples of optical scanners include:**
 - **Optical Mark Recognition Scanners (OMR)**
 - **Optical Bar Recognition (OBR)**
 - **Optical Character Recognition (OCR)**

Optical Mark Recognition Scanners (OMR)

- OMR detects marks made on a piece of paper using ink or a soft pencil by passing an infra red beam over them.



- `OMRs are used to mark multiple choice questions, analyzing responses to structured questionnaires and selecting correct number combinations from lottery tickets.

Optical Bar Recognition (OBR)



- Are used to capture data coded as lines of varying thickness known as barcodes.
- Barcodes hold manufacturers details and the product code.
- Type of scanner used to read these bars is called Wand or Laser scanner.

Optical Character Recognition (OCR)



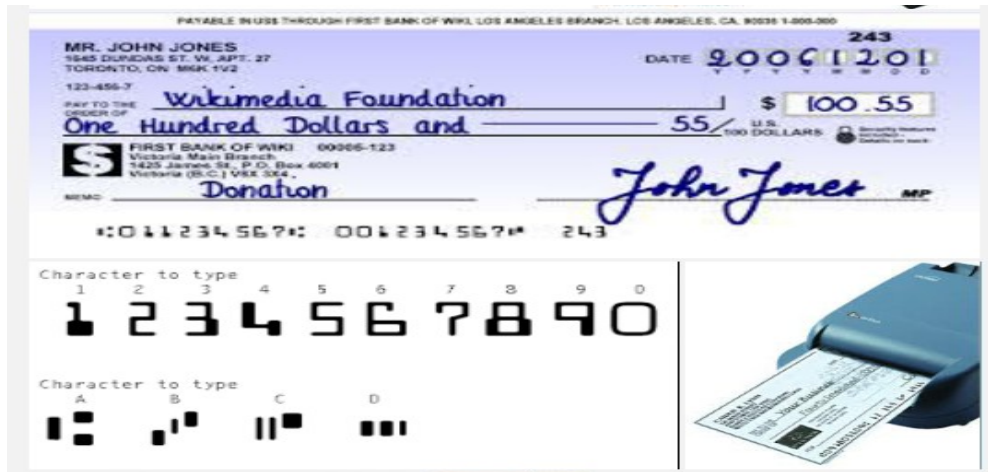
- An OCR scanner is used to read typewritten, computer printed or handwritten characters and transforms the images into a soft copy that can be manipulated using a word processor.

2. MAGNETIC SCANNERS

- Are used to capture data written using magnetic ink or coded onto a magnetic strip.
- Common examples of magnetic scanners are :
 - Magnetic Ink Character Recognition (MICR)
 - Card Readers

Magnetic Ink Character Recognition (MICR)

- used to read characters written using magnetic ink.
- An example of an MICR is the cheque reader used in banks.



Other Input Technologies

- **Touch Screen:** Lets the user to touch the screen using a finger or a stylus. In order to write or select an item.
- **Voice Input:** A microphone is used to enter data in form of speech into the computer.
- **Interactive Whiteboard:** Also known as smart board. Is a large interactive display that connects to a computer.

SUMMARY:

Optical Scanners- capture data using optical or light technology

Magnetic Scanners- capture data written using magnetic ink or coded onto a magnetic strip

Output Devices

- Output devices are peripheral devices that a computer uses to give out information produced after the processing operations.
- These devices can be classified into two categories namely:
 - Soft Copy output devices
 - Hard Copy output devices

Soft Copy Output Devices

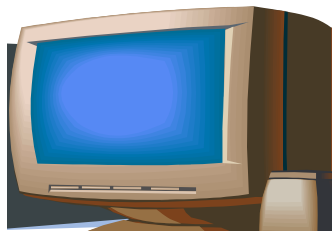
- Soft copy refers to intangible output that can be seen or heard such as screen display and sound.
- Examples of such devices include :
 - Monitors
 - LCD Projectors
 - Speakers

Monitors

- A monitor also known as Visual Display Unit (VDU) or the screen ,is used to display information in the form of text, pictures and video enabling the user to monitor what is going on in the computer.
- There are three common types of monitors namely :
 - Cathode Ray Tube (CRT)
 - Liquid Crystal Display (LCD)
 - Gas Plasma Display (GPD)

Cathode Ray Tube Monitors

- A CRT Monitor consists of a long glass tube with an electron gun on one end and screen on the other.
- The electron gun shoots electrons to illuminate the screen and is coated with tiny phosphorus dots.



Liquid Crystal Display (LCD) Monitor

- An LCD monitor is made of special liquid crystals.
- Unlike CRTs, LCDs are less bulky, consume less power and have little strain effect on eyes.

Gas Plasma Display

- Gas plasma displays resembles LCDs only that they make use of gas instead of liquid crystals.
- They contain millions of pixels that are illuminated by charged neon gas.
- Unlike LCDs, images displayed on gas plasma do not suffer from angle distortion.

Other Output Devices

- **Sound output devices** produce sounds such as beeps, audio or digital.

Data Projectors

- Are used to display output from a computer onto a plain white screen like a wall or a white board.

Light Emitting Diodes

- LEDs are indicators that display light when an electric current is passed through them.

Hard Copy Output Devices

- Refers to tangible output that can be felt such as paper.
- Examples of hard copy output devices include:
 - Printers
 - Plotters
 - Facsimile (Fax)

Printers

- Printers are primarily used to produce information on a piece of paper.
- The quality of a hard copy depends on the printers printing mechanism.
- Printers are classified according to different printing mechanism.

- The two categories are:
 - Impact Printers
 - Non-Impact Printers

Impact Printers

- Impact Printers print using striking mechanism.
- This means that they strike the paper in order to form an imprint on it.
- Two examples of impact technologies are:
 - Dot Matrix Printers
 - Daisy Wheel Printers

Non Impact Printers

- Are faster and more quiet than the impact printers.
- They print using ink, thermal, or laser mechanisms.
- The four types of non impact printers are:
 - Inkjet
 - Thermal
 - Laser
 - Photo

Factors to consider when purchasing a Printer

- **Print Quality** -Laser printers are good for printing official documents.
- **Initial Cost** - Laser and Thermal printers are expensive compared to inkjet printers.
- **Running Cost** - Cost of maintaining an inkjet is higher than that of maintaining laser printers.
- **Speed** - The speed of a printer is measured in Pages per Minute.
- **Colour Printing** - Most printers support black and colour printing. However colour printers are more expensive.

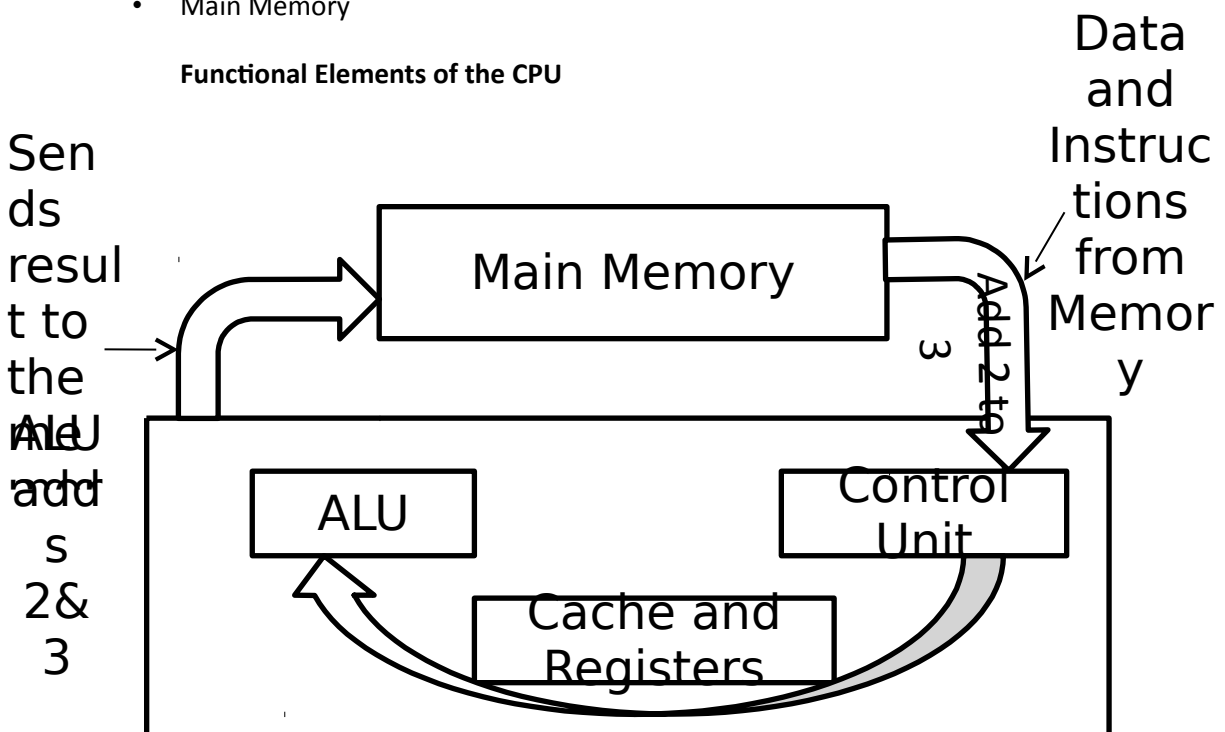
Plotters

- Is a large type of hard copy output device.
- Plotters are mostly used for printing geographical ,architectural and engineering drawings e.g. maps, advertisement posters to be placed on bill boards, machine parts etc.

The Central Processing Unit (CPU)

- The Central Processing Unit (CPU) also known as the *processor* is the most important component of the computer.
- It is usually regarded as the brain of the computer because all processing activities are carried out inside the processor.
- In microcomputers, the CPU is housed inside the system unit.
- It is mounted on a circuit board known as the *motherboard* or the *system board*.
- The CPU consists of three functional elements namely:
 - Control Unit
 - Arithmetic and Logic Unit (ALU)
 - Main Memory

Functional Elements of the CPU



Control Unit

- It coordinates all processing activities in the CPU i.e input, storage and output operations.
- It determines which operation or instruction is to be executed next.
- To coordinate these activities, the Control Unit uses a **system clock**. System Clock sends its **electric signals** as its means of communication.
- The number of **pulses per second** determines the **speed** of a microprocessor.
- The faster the clock pulses, the faster the CPU, hence the faster the computer can process data.

Arithmetic and Logic Unit (ALU)

- The arithmetic and Logic unit is a unit of central processing unit where all arithmetic and logical operations are carried out.
- The basic arithmetic operations include; addition, subtraction, multiplication and division.
- Logic operations are based on the computers capacity to compare two or more values.
- In order for the ALU to be able to process data, it has special locations called **registers**, which hold the data just before and after processing.

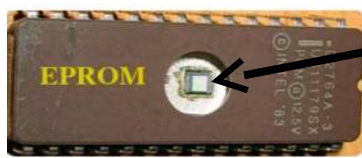
Main Memory

- Main memory also known as primary storage is a type of storage that is directly accessible by the processor.
- Computer memory can be classified into:
 - Read Only Memory (ROM)
 - Random Access Memory (RAM)

Read Only Memory (ROM)

- Read Only Memory is used to store programmed instructions and data permanently or semi-permanently.
- Data and instruction stored in ROM are those which remain unchanged for long periods of time e.g. POST instructions.

- Depending on permanence of the instructions or data written on it, there are four types of Read Only Memory namely;
1. **Mask Read Only Memory:** Once the content is written on it by the manufacturer ,it cannot be changed.
 2. **Programmable Read Only Memory (PROM):** This allows the user to alter it only once after the content is written on it.
 3. **Erasable Programmable Read Only Memory (EPROM):** It has a transparent quartz window through which its content can be erased by exposing it to ultra violet (UV) light and then reprogrammed for another use.



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4. **Electrically Erasable Programmable Read Only Memory (EEPROM):** This type of ROM can be erased and reprogrammed using electricity.

Characteristics of ROM

- One can only read its contents but cannot write on it unless it is a special type of ROM.
- It is non-Volatile i.e. its content is not lost when the computer is switched off.
- Stores permanent or semi permanent instructions.

Random Access Memory (RAM)

- It is also known as working storage.
- It is used to hold instructions and data needed by the currently running applications.
- The information in RAM is continually read, changed and removed.

Characteristics of RAM

- Data can be read and written on it.
- RAM is a temporary storage.
- It is volatile meaning its content disappears when the computer is switched off.
- Its content is user defined i.e. the user dictates what is to be contained in the RAM.

Types of RAM

- **Static RAM:** It is a fast type of memory located inside a microprocessor.
- **Dynamic RAM:** Is a relatively slower type of RAM compared to SRAM.

Overall

Functional Organization of the CPU

- The Arithmetic and logic unit, the control unit and the main memory use electrical pathways or links referred to as buses. There are three types of Buses namely:
- **Control Bus:**
 - is for all timing and controlling functions sent by the control unit to other parts of the system.
- **Address Bus:**
 - is used to locate the storage position in memory where the next instruction data to be processed is held.
- **Data Bus:**
 - is where the actual data transfer takes place.

NB: Bus is a pathway.

Secondary storage Devices and Media

- Secondary storage also called Auxiliary storage are devices that provide alternative long term storage for programs, data and information.
- They are regarded to as secondary because unlike primary storage, they are not directly accessible by the CPU.

- Secondary storage devices can be classified according to:
 - Portability as removable and Fixed
 - Technology used to store and retrieve data as magnetic, optical, magneto-optical and solid state.

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Removable Storage

- Removable storage media are those that are not housed inside the computer.
- Data is read and written into the media using a device known as a drive.
- Examples of removable storage include:
 - Magnetic Tapes
 - Floppy Disks
 - Optical Disks
 - Solid State devices.

Optical Storage Media

- Optical storage media are so called because data is written and read from them using a laser beam.
- A laser beam is a very strong concentrated light.
- The reason why optical storage media are used :
 - They store very large volumes of data.

- Data stored in them is more stable and more permanent than the magnetic media.

Compact Disks (CD)

- Compact Disks hold large quantities of data and information
- One disk can hold as much as 700MB.
- They are mostly used to store data and information that requires a lot of space such as video clips, software, sounds etc.
- Currently compact disks are available in three forms namely:
- **Compact Disk Read only Memory (CD-ROM):** They contain data that can only be read but cannot be written on.
- **Compact Disk Recordable (CD-R):** Once data is burned on a CD-R , it becomes read only.
- **CD-ROMs** and **CD-Rs** are referred to as **Write Once Read Many (WORM)**. Data is only recorded once but can be read as many times as possible.

Compact Disk-Rewritable (CD-RW): Unlike the CD-Rs, these type of compact disks allows the user to record, erase and rewrite new information

Digital Versatile Disks

- DVD also known as Digital Video Disk resemble compact disk in every aspect.
- The only difference is that they have a higher storage capacity of up to 17 GB.
- They are suitable for recording motion pictures such as videos because they offer better sound and picture quality.

Optical Card

- An optical card resembles an MICR card but data is read and written optically on a stripe.
- They are mostly used in banking.

Optical Tape

Similar to magnetic tape only that data is stored on it by using optical technology

Solid State Storage Media

- Is a non-volatile storage that employs integrated circuits.
- They are referred to as solid state because they do not have movable parts

- Examples include:
 - Memory Sticks
 - Floppy Disk
 - Flash Disks
 - Solid State devices.

Fixed Storage Media

- Traditionally the hard disk was mounted inside the computer.
- However due to advancement in technology nowadays hard disk are removable.

Computer Software

- In order for a computer to process data, it is guided by a set of instructions known as programs.
- The term software and program are used interchangeably.
- The term software is however not only used to mean programs but also the documentation associated with it such as user guides.
- There are three major ways of classifying computer software;

Classification of Computer Software

- According to purpose
- According to Acquisition
- According to End user License (EUL)

Classification According to purpose

- Computer software may primarily be designed to manage the hardware resources or to help the user accomplish specific tasks.
- Therefore software may be classified as either :
 - System Software
 - Application Software

System Software

- System software performs a variety of fundamental operations that avails computer resources to the user. These functions include:
 - Booting the computer and making sure that all the hardware elements are working properly.
 - Performing operations such as **retrieving, loading, executing** and **storing** application programs.
 - Storing and retrieving files.
 - Performing a variety of system utility functions.
- System software can further be subdivided into four sub- categories namely:
 - Operating systems
 - Firmware
 - Utility Software
 - Networking Software

Operating Systems

- An operating system consists of a set of complex programs that work together to control the execution of user programs called applications and acts as a go between (interface) between the applications and the computer hardware.
- It manages input, output and storage operations in a computer.
- Examples of common operating systems include:
 - Microsoft Windows (2000, XP, Vista, 7, 8)
 - UNIX

- Linux
- Macintosh (MacOS)

Firmware

- It is also referred to as stored logic.

It is a combination of both the software and hardware recorded permanently on electronic chips.

$$\text{Software} + \text{Hardware} = \text{Firmware}$$

Utility Software

- Is a special program that performs commonly used services that make certain aspects of computing to go on smoothly.
- Such services include:
 - Copying
 - Sorting
 - merging
 - File handling
 - Disk Management

Networking Software

-Used to establish communication between two or more computers by linking them using a communication channel like cables to create a computer network.

- Networking software enables the exchange of data in a network as well as providing data security.

Application Software

- Application software, also known as application packages are programs that are designed to help the user accomplish specific tasks.
- The table below shows examples and uses of common application packages.

Information Communication and Technology

Classification According to Acquisition

- Software can be classified according to acquisition as:
 - In –house developed
 - Vendor off-the-shelf software

In-House Developed Software

- These are programs that are uniquely designed and tailored to meet a particular users needs.
- These programs are not available in the shops and once developed for one company they may not address the needs of other users

Vendor off-the –shelf Software

- These software are developed by software engineers, packaged and then made available for purchase through a vendor, a distributor or directly from the developer.
- A developer may bundle more than one but closely related software into one package to form a suite or integrated software;
- Examples of suites are:

- Microsoft Office Suite
- Lotus Suite

In-House Developed Software

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Advantages of vendor off the shelf over the in-house developed

- Easily installed and run.
- Cheaper
- Available for almost any task.
- Minor or no errors since they are thoroughly tested.
- They can easily be modified to meet users needs.

Disadvantages of Vendor off the shelf

- May lack some features required by the user.

- They may have some features not needed by the user which may take extra storage.
- They may require the user to change processes and hardware for compatibility which may in turn be expensive.

Classification According to End-User License

- Software may also be classified according to End user License as:
 - **Non-Proprietary/ Open Source-** are softwares whose source code is freely available to the users. The users are encouraged to use, modify and distribute the modified product.
 - **Proprietary-**Are softwares whose source code is hidden from the users. Modifications are only made by the software manufacturer.
 - **Freeware-** Are software products that are freely made available to the user. However ,beware of freeware because some may be malicious software.

Criteria for selecting a computer system

- Before purchasing a computer, hardware or software resources, it is advisable to do a requirement analysis of all the requirements necessary in order to balance between cost and user needs.
- Requirements analysis involve, identifying user needs and software and hardware evaluation.

Hardware Considerations

- Microprocessor type and speed
- Warranty
- Cost
- Upgradability and Compatibility
- Portability
- Special user Needs
- Monitor
- Multimedia Capability

- Available Ports
- Wireless Connectivity
- System Unit Form Factor

Software Considerations

- Authenticity
- Documentation
- User Needs
- Reliability and Security
- User Friendliness
- Cost
- Compatibility and System Configuration
- Portability

WORD PROCESSORS

- A word processor is application software used to create, save, edit, format and print text documents.
- Word processing is the act of creating, saving, editing, formatting and printing text documents.
- Examples of word processors include:
 - Microsoft word
 - Corel WordPerfect
 - Word Star
 - Lotus WordPro

Advantages of a word processor

- Document can be stored for future reference.
- Typing using a word processor is easier and more efficient because some actions are automated e.g. *word wrap*, *auto correct* features.
- Word processors have special editing tools such as spelling and *grammar checkers* and *thesaurus*.
- Have superior document *formatting* features.
- In word processing one can print many copies at once.

WORD PROCESSING FEATURES

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WORD PROCESSING FEATURES

- Allow the user to create a file, save it, and open it again and again for reuse, editing and formatting.
- All word processors have a spelling checker, thesaurus and grammar checker.
- They provide for the use of headers and footers, indexing, footnotes and references.

Most word processors have ability to create and import tables, text and graphics from other programs.

Factors to consider when choosing a Word Processor

- The type of operating system-Most microcomputers currently run on windows based operating systems such as *Microsoft Windows* this means one should consider acquiring a graphical user interface based word processor.
- Its user friendliness i.e. ease of use.
- Its formatting and editing features. They should be good and varied.

Terms used in Word processing

Word Wrap: The feature that automatically moves a word or cursor to the beginning of the next line.

Scrolling: Is the vertical or horizontal movement of text document on the screen.

- **Thesaurus:** Is an editing tool that provides the user with *synonyms* and *antonyms*.
- **Auto complete:** Is a feature that automatically displays a complete word when the user types the first few characters of the word.
- **Auto correct:** Is a feature that automatically detects wrongly spelt or capitalized words and replaces them with the correct word.

Parts of a window

Title Bar: Displays the title of the currently running application or task.

On its right are the minimize, *restore/maximize* and the *close* buttons.

The minimize button reduces a window to become a button on the task bar.

The *restore/maximize* stretches the window to cover the entire desktop or restore it to its original size respectively.

The *close button* is used to close and exit a window.

Menu Bar: Provides a drop down list of commands that one can use to perform a task.

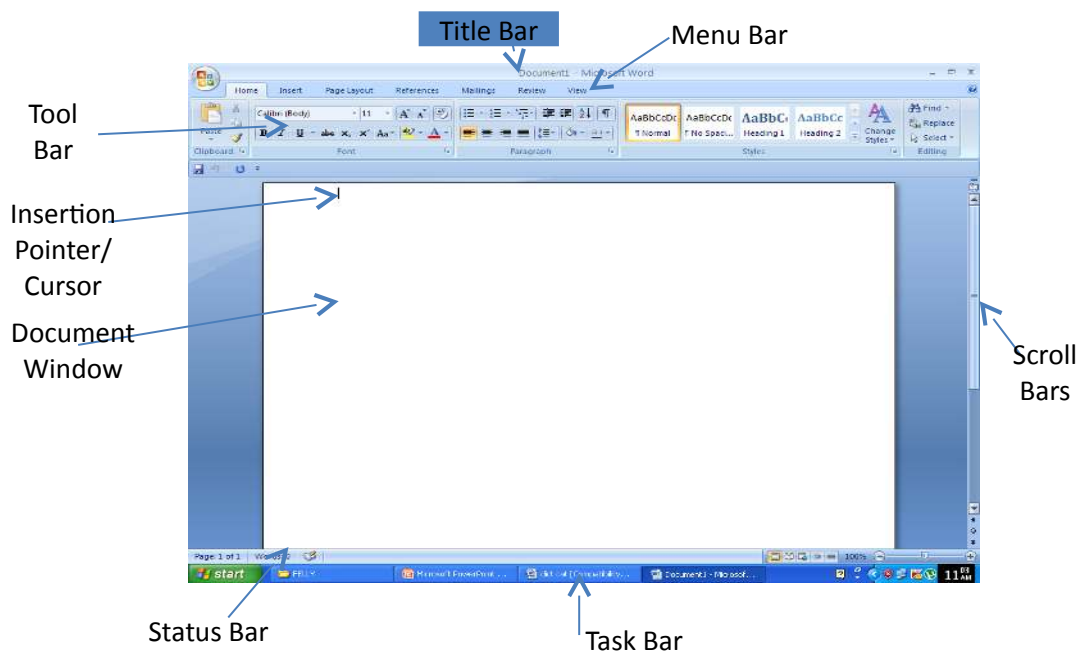
Toolbars: These are rows of buttons or icons that represent commands. These command buttons are normally shortcuts.

- **Work area/Document window:** Is the working area where you can enter text or graphical objects.
- **Status Bar:** Is an interactive strip at the bottom of the screen that acts as a communication link between the user and the program.

Selecting with the Keyboard

- **Select one word**-Move the pointer to the beginning or end of the word then press Shift+Ctrl+Right or Left arrow.
- **Select one line:** Shift + Up or Shift +Down arrow key or Shift +Home or end keys respectively.
- **Select an entire page:** Press Shift +Page Down or Shift +Page up.
- **Select an entire document:**Ctrl +A or Shift +Ctrl +End.

Parts of the Microsoft Word Window



SPREADSHEETS

- An electronic spreadsheet simply referred to as a spreadsheet
- **Spreadsheet** is application software used to calculate, organize and analyze numerical data.
- It consists of rows and columns similar to the ones of the manual spreadsheet.

Examples of Electronic spreadsheets are:

- Microsoft Excel

- Open Office Calc
- Lotus 1-2-3
- Corel Quattro Pro

Advantages of Electronic Spreadsheet

- Offers a large virtual sheet for data entry and manipulation.
- Has better document formatting capabilities.
- Has inbuilt formula called *functions* for manipulating mathematical data.
- Utilizes the powerful aspects of the computer like speed, accuracy and efficiency to enable the user to quickly accomplish his/her tasks.
- Utilizes the large storage space on computer storage devices to save and retrieve documents.
- Enables the user to produce neat work because the traditional paper, pencil, rubber and calculator are put aside.
- Automatically adjusts the result of a formula if the values in a worksheet are changed. This is called the *automatic recalculation* feature.

Automatic Recalculation

- *is a feature that* automatically adjusts the result of a formula **if** the values in a worksheet are changed.

Components of a Spreadsheet

A spreadsheet has three main components mainly:

- **Worksheet-** is the work area made up of rows and columns where data is entered. Each row is labeled using a number while each column is labeled with a letter.
- **Database-** is a collection of related data items organized so as to provide a consistent and controlled access to items.
- **Graph/Charts-** is a pictorial representation of the base data on a worksheet. Most spreadsheets refer to graphs as charts. A chart enables the user to present complex data elements from a worksheet in a simple and easy to understand format. Examples of charts are: Line charts, Bar charts.

Application areas of a Spreadsheet

1. Statistical Analysis

Spreadsheets provide a set of data analysis tools that can be used to perform simple analysis such as computing mean, mode, and standard deviation as well as complex statistical analysis.

2. Accounting

Accountant find electronic spreadsheets useful tools for analyzing financial transactions such as computing totals, consolidating financial records as well as predicting future business trends.

3. Mathematical and Scientific

Spreadsheets are used to solve mathematical and scientific problems such as arithmetic and trigonometric.

4. Forecasting

A feature known as "What if "analysis is used to make future predictions. Forecasting is an important function used for predicting economic trends budgeting, stock portfolio analysis, cost analysis and cash flow analysis.

Features of Microsoft Excel Screen

1. Formula Bar

Is where a cell entry or a formula is displayed before being entered into the selected cell.

2. Worksheets

Is the work area made up of rows and columns where data is entered. The intersection of rows and columns forms cells.

3. Cells

A cell is an intersection between a row and a column. A column is a vertical arrangement of cells while a row is the horizontal arrangement of cells.

4. Active cell pointer

A cell pointer indicates the current active cell.

5. Cell Address

Is the combination of a column header and a row header that indicates the location of a specific cell.

6. Worksheet Tabs

Are located at the bottom of a spreadsheet. They show the number of worksheets in the workbook. A workbook is a spreadsheet file that consists of one or more related worksheets.

Shift + F11.....inserts a new worksheet.

CELL DATA ENTRIES

Entries into a worksheet can be classified into four categories namely:

1.Labels

Any text or alphanumeric characters entered in a cell are viewed as labels by the spreadsheet program. Labels are used as row or column headings usually to describe the contents of the row or column e.g. if a column has names of people, the column header can be NAMES.

2. Values

These are numbers that can be manipulated mathematically. They may include currency, date, numbers (0-9), special symbols or text that can be manipulated mathematically by the spreadsheet.

3. Formulae

Formulae are mathematical expressions that create a relationship between cells to return a new value.eg the formula=B3+D4 adds the contents of B3 and D4 and returns the sum value in the current cell.

4. Functions

These are inbuilt predefined formulae that the user can quickly use instead of having to create a new one each time a calculation has to be carried out.eg to add contents of cells say B3 to E3
type=SUM(B3:E3)

CELL REFERENCING

There are three types of cell referencing used when creating formulae and manipulating cell content. These are:

1. Relative Referencing

When you copy a formula that contains cell references, say **A3**, the references adjust to their new location e.g. if **C1** contains a formula **=A1+B1** is copied to **C2**, the references change to **=A2+B2**.

2. Absolute Referencing

The reference is made to a specific address and does not change even if the formulae is copied to another cell. It is made by placing a dollar sign before the reference e.g. **=A\$2**. If the result in **C1** is 170, if absolute reference is used, 170 will be copied to **C2**.

3. Mixed Cell Referencing

This is a combination of relative and absolute reference. e.g. **\$A3** or **A\$3**. In the first case the column reference is absolute while the row is relative, in the second case the column reference is relative while the row reference is absolute.

Functions

- A function must start with an (=) sign followed by the *function name* and the *arguments*.
- Arguments are numeric, logical values or text enclosed in parenthesis/brackets.
- For example, in **=SUM (A3:F3)**, SUM is the function that adds the range A3 to F3 which is the argument.

TYPES OF FUNCTIONS

1. Mathematical functions
2. Statistical Functions
3. Logical Functions
4. Date and Time Functions

Mathematical Functions

- **SUM()** Adds all the values in the selected range of cells. e.g. if A3, B3, and C3 contain 20, 50 and 80 respectively, **=SUM (A3:C3)** returns 150.
- **ROUND ()** Rounds a number to a specified number of decimal places. Zero rounds off the number to the nearest integer. e.g. **=ROUND (49.769,1)** returns 49.8 while **=ROUND (49.769,0)** returns 50.

- SUMIF() Conditionally adds the specified cells according to the set criteria. =SUMIF(A3:A10,">=1000") returns a value if ,and only if the sum is greater than or equal to 1000.
- PRODUCT() Multiplies all the values in the argument. e.g. =PRODUCT(40,3,2) returns 240.

Statistical Functions

- AVERAGE() Returns the arithmetic mean of its arguments. e.g. if A3,B3,and C3 contains 20,50 and 80 respectively,=AVERAGE(A3:C3) returns 75.
- COUNT() Counts the number of cells that contain numbers within a range. e.g. =COUNT(A3:C3) returns 3.
- COUNT IF () Conditionally counts the number of cells within a range that meets a given condition. e.g. if A3,B3,C3,D3,and E3 contains 20,50,80,60 and 45 respectively then=COUNTIF(A3:E3,">=50") returns 3.
- MAX() Returns the largest value in a set of values or within a range. e.g. =MAX(A3:E3) for the values above returns 80.
- MIN() Returns the smallest value in a set of values or within a range. e.g.=MIN(A3:E3) for the values above returns 20.
- RANK() Returns the position or rank of a number from a list of values. e.g. =RANK(A2,\$A\$3:\$A\$8,1) returns the position of A2 as if the list was sorted in ascending order.

Logical Functions

- ➡ **IF()** Returns a value or label if a condition you specify is evaluated as TRUE and another is evaluated to FALSE. e.g. if A3,B3,C3,D3,and E3 contains a set of marks 35,50,80,60 and 45 grades are to be awarded as follows:

80 to 100	A
60 to 79	B
40 to 59	C
Below 40	Fail,

To assign a grade use,

=IF(A3>=80,"A",IF(A3>=60,"B",IF(A3>=40,"C","FAIL")))

AND() Returns true if all its arguments are true and false if any argument is false. e.g. =AND(3+2=5,2+2=4) returns True.

OR() Returns true if any of its arguments is true or false if both arguments are false. e.g.=OR(3+2=7,2+2=4) returns True.

NOT() Negates the unary operand. e.g. =NOT(3+2=5) returns False.

Date and Time Functions

- **TODAY ()** Returns a number that represents today's date. The function takes no arguments.
- **NOW()** Returns the current date and time formatted as date and time. It takes no arguments.
- **DATE()** Returns a serial number that represents a particular date. Microsoft Excel uses year 1900 serial number 1. For example, = DATE (111,1,4) returns January 4, 2011. Year=(1900+111), month =1, day=4.
- **HOUR(), MINUTE() or SECOND()** This Function returns the current hour as number 0 to 23, minute from 0 to 59 and seconds from 0 to 59 respectively.

Terms Used In Charts

- **Title**

Is a heading that shows clearly what a chart represents.

- **Legend**

Is equivalent to a key and shows what each color or pattern of data series represent.

- **Data Value**

Are representations of data on a chart. Data can be labelled either using a value or text label.

➡ **Data series**

Displays the cell or range that contains the name of the selected data series. The series are normally separated by the \$ (dollar) sign.

DATABASES

Database is a collection of structured and related data items organized to provide a consistent and controlled access to the items.

- To create and manage databases, we use Database Management System (DBMS) Software.

Database Management System Software facilitates the following:

1. Creation of database
2. Organization of database
3. Maintenance of databases.

- **Examples of Databases management software include:**

- Microsoft Access
- Microsoft SQL Server
- Oracle
- FoxPro
- DbaseIV
- Lotus Approach
- MySQL

Functions of Database Management Software

- Allow the user to add or delete records.
- Update or modify existing records.
- Acts as interface between a database and other application programs.
- Organize data for easy access, retrieval and manipulation of records.

- Ensure security for the data in the database by safeguarding it against unauthorized access and corruption (damage).
- Keep statistics of data items in a database.

Database Models

Databases can be classified according to the method used to organize data.

The main database models are:

- Flat file
- Hierarchical
- Network
- Relational
- Object oriented model

Cont. of Data Field Types

Text

Use for text or combinations of text and numbers, such as addresses, or for numbers that do not require calculations, such as phone numbers, part numbers, or postal codes.

Stores up to 255 characters. The Field Size property controls the maximum number of characters that can be entered.

Memo

Use for lengthy text and numbers, such as notes or descriptions.

Stores up to 65,536 characters.

Number

Use for data to be included in mathematical calculations, except calculations involving money (use Currency type).

Stores 1, 2, 4, or 8 bytes; stores 16 bytes. The Field Size property defines the specific Number type.

Features of a Database Management Software

Most database software contains a collection of features that provides the users with a means to manipulate data in a database.

These features include:

- Tables
- Queries
- Reports Generators
- Form Interface
- Pages
- Macros

Data organization in a Database

One of the functions of a database system is to organize data for easy access, retrieval and manipulation.

Data is organized from the simplest form called a *field* to a very complex structure called a *database*.

Fields: A field is a character or a logical combination of characters that represent a data item.eg in a class list the student name is a field.

Records: A record is a collection of related fields that represent a single entity.

Tables: A table is a collection of related records.

Database: This is the highest in data organization hierarchy that holds all related files or tables.

Description of Field Data Types

A field's data type can be defined as a set of qualities that applies to all the values contained in the field and that determines what kind of data those values can be.

The type of data to be used in a database must be clearly defined for the purpose of manipulation and storage.

The data types allowed in Ms. Access include:

Date/Time

Use for dates and times. Stores 8 bytes.

Currency

Use for currency values and to prevent rounding off during calculations. Stores 8 bytes.

AutoNumber

Use for unique sequential (incrementing by 1) or random numbers that are automatically inserted when a record is added. Stores 4 bytes; stores 16 bytes.

Yes/No

Used for data that can be only one of two possible values, such as Yes/No, True/False, On/Off. Null values are not allowed. Stores 1 bit.

OLE Object

Object Linking and Embedding. Used for OLE objects (such as Microsoft Word documents, Microsoft Excel spreadsheets, pictures, sounds, or other binary data)

Stores up to 1 gigabyte (limited by disk space).

Hyperlink

Use for hyperlinks. A hyperlink can be a UNC path or a URL.

Stores up to 64,000 characters.

Lookup Wizard

Setting a data type as a look up wizard starts the Lookup Wizard, which helps to create a Lookup field. A Lookup field displays either a list of values that is retrieved from a table or query, or it displays a static set of values that you specified when you created the field.

Field Properties

Field properties depend on the type of field selected.

Field Size: Allows the user to set the number of characters in a field instead of the default 50.

Format: Determines how information appears on the screen when printed.eg you can format a number to scientific, currency, percentage etc.

Decimal Places: For number and currency fields, you can specify the number of decimal places.

Input Mask: Input mask automatically formats the field entry into a specified format.

Caption: Is a more descriptive name for a field to be used in a table or a form display.eg the caption for *StuName* could be student name.

Default Value: Is a value that appears automatically in the datasheet or form if nothing is entered by the user to change it.

Validation Rule: Logical expression that restricts the values to be entered in a field.

Validation Text: Is a message that appears once the validation rule is violated.

Required: Determines if an entry must be made in the field before you proceed to the next field or record.

Allow Zero Length: This allows the user to proceed without making any entry in the field set as Zero length.

Indexed: An index facilitates the organization of records for easy search. A primary key is an example of an index set to *No duplicates*.

Primary Key and Indexes

Primary key - is an index that uniquely identifies each record stored in the table.

Functions of a Primary key:

- Prevents the user from making null or double entries into a table
- To order records, and control redundancy.

An index is a key(s) used to speed up searching and sorting records in a table.

A foreign key is a field in a relational table that matches the primary key column of another table. The foreign key can be used to cross-reference tables.

Relationships

There are different types of relationships that can be established between tables.

Types of Relationships

- **One to One**

This is where one record from a Table A can only be related to one record from Table B. eg One person has only one National ID Number.

- **One to Many**

Is the situation where one record from a Table A is related to at least one from Table B. e.g Instructor/Student.

- **Many to Many**

One record in either Table A or Table B, can relate to many records in the other table. eg customers purchasing many products and products can be purchased by many customers.

- **Indeterminate**

This is where none of the related fields is a primary key. To rectify this set at least set one of the fields as the primary key.

Referential Integrity

This is a security feature that helps keep track of changes especially to the data in the join fields.

Join Lines: A join determines how to connect the information together from many tables so that Ms. Access knows how to connect the information.

MICROSOFT POWERPOINT

Microsoft PowerPoint is a graphics presentation program. It can be used for designing presentations quickly.

PRESENTATIONS.

PowerPoint is most useful for creating formal presentations. PowerPoint can help you produce professional looking presentations that can be printed on Over Head Projectors transparencies, viewed on a computer display or produced as a slide, together with printed notes for your audience to take away for reference.

Presentations can be used in certain situations to convey information more effectively. Examples of such situations include;

- 1.** Marketing or internal company presentations.
- 2.** To deliver company results to a shareholder meeting.
- 3.** To report sales figures.
- 4.** To launch a new product into the market.
- 5.** To introduce/induct new employees into a company operations.
- 6.** Advertising in the media, trade fairs or exhibitions.

With PowerPoint, you can create overhead slides, Speakers notes, audience handouts, and an outline all in one single presentation file. PowerPoint has powerful wizards to help you create and organize your presentation systematically.

Microsoft PowerPoint is also useful as an aid to Desktop Publishing.

Starting Microsoft PowerPoint.

Purpose.

✓ You can start PowerPoint whenever you want to create or run a presentation.

1. Click **Start**, point to **Programs**, then select **Microsoft PowerPoint** from the resulting pop-up menu.
2. The **PowerPoint** Startup dialog box appears giving you four (4) options from which you must choose one in order to proceed with any other PowerPoint features or functions.

◆ *The AutoContent Wizard:*

This option starts the AutoContent Wizard, which is a sequence of screens offering you with alternations.

The wizard allows you to select the type of presentation that you would like to create, and then prompts you for a title for your presentation as well as information about the topic that you have chosen.

PowerPoint then provides a basic outline to guide you in organizing your content into a professional presentation.

◆ *Design Template:*

This option gives you a list of presentation designs; presentations and other templates that have predefined slide colors and text styles.

◆ *Blank Presentation:*

This option brings up the **New Slide** dialog box, which has 24 blank slide layouts. The slides are pre-designed.

After choosing a layout, the user fills in the rest of the content.

◆ *Open an Existing Presentation:*

Use this option to choose an existing PowerPoint presentation that had already been saved in the computer.

Using the Different PowerPoint Views.

Purpose.

✓ PowerPoint has six views that help you to view or work on your presentations in different ways.

The View buttons are usually located at the bottom left of the presentation window. The different views are: -

- 1). Normal view.
- 2). Slide view.
- 3). Slide Sorter view.
- 4). Outline view.
- 5). Notes Pages view.
- 6). Slide show.

The main views you use in PowerPoint are Normal view, Slide view, and Slide Sorter view.

To switch between the different views, click on the View buttons at the bottom of the screen,

-OR-

Select one of the options from the **View** menu.

Normal View:

Normal View simultaneously displays the Slide, Outline, and Notes views in their own, allowing you to see everything at once.

These panes let you work on all aspects of your presentation in one place. You can adjust the size of the different panes by dragging the pane border.

In the Outline pane, you can organize and develop the content of your presentation. You can type all the text of your presentation and rearrange bullet points, paragraphs, and slides.

In the Slide pane, you can see how your text looks on each slide. You can add graphics, movies, and sounds create hyperlinks and add animations to individual slides.

In the Notes pane, you can add your speaker notes or information you want to share with the audience.

Slide View:

Slide view is the default view when you open or create a presentation.

In Slide view, you can work with the text in your presentation one slide at a time.

Outline View:

Outline view allows you to see all the titles, text, etc of the entire presentation in a single window.

In the Outline view, the text appears exactly the way it appears when you work with Microsoft Word, i.e., you can see how your main points flow from slide to slide.

In Outline view, each slide is numbered. A Slide icon appears to the left of each slide's title. The text below each Slide title is indented. This is the same text contained in that particular slide.

In Outline view, you can edit the text, add and delete slides, move text between slides and change the order of the slides themselves.

You can type in, view or edit the text for all your slides at once. To type in text, position the cursor, type in the text, and then press the ENTER key.

Slide Sorter View:

Slide Sorter View allows you to:

- Easily add, delete and move slides.
- Preview the entire presentation.
- Organize or rearrange the order of the slides in your presentation.
- Preview animations on multiple slides by selecting the slides you want to preview, and then click **Animation Preview** on the **Slide Show** menu.

Use the buttons on the **Slide Sorter** toolbar to set timings for a slide show and transition effects when moving from slide to slide.

Notes Page View:

This view enables one to create/type Speaker's notes. These are the notes that a presenter uses as 'cue cards' or reminders during a presentation.

A miniature slide appears above the speaker's notes.

you can also print a copy of your notes for reference.

Slide Show:

Slide Show view displays slides on the entire screen the way the audience will view the final presentation.

Presentation Tips.

Purpose.

- ✓ An effective presentation is well prepared and carefully planned.
An effective presentation is more than just slides that contain text and shapes. It often combines text, shapes, charts, colour, and sometimes sound or movies to produce a successful communication tool. This is due to the fact that, most people retain 40% - 50% more information when they see and hear simultaneously rather than from just listening.

Characteristics of an Effective presentation.

Effective presentations have some common characteristics, which should improve communication between the speaker and the audience.

Simplicity.

Good presentations are simple and to the point. They are not long-winded and boring or complicated.

They are easy for the speaker to present and easy for the audience to understand.

A long-winded and complex presentation usually ends up confusing the audience contrary to expectations.

Visuals.

Good presentations make use of a lot of visuals, i.e., graphics, pictures, ClipArt, charts, and so on.

These generally serve the purpose of:

- 1). Attracting attention to the pertinent (important/relevant) points of a presentation.
- 2). Helping relieve stress and 'eye fatigue' from too much reading.
- 3). Making your presentation more interesting.
- 4). A visual can tell the audience much more about the subject matter than a slide full of text. In other words, "A picture says more than a thousand words".

Colour.

A good presentation usually has a lot of colour. Compared to a black and white presentation, a colour presentation attracts a lot of attention.

However, the use of colour must be judicious (sensible / well judged), otherwise, the presentation will end up being overly 'colourful' and too flashy.

Creating an Effective Presentation.

Having determined the characteristics of an effective presentation, the next logical step is to create the presentation using the tools covered in this package.

There are THREE steps that need to be followed in order to create an effective presentation.

1. Planning.
2. Preparing.
3. Execution.

